

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 169

THE CHEWING DOCTRINE

MOUTH-FEEDERS, DOMES, CASSETTES

Chew or lose them

button-dosed, deadman-sealed, one piece at a time



paste is the failure mode, not the menu

FOUR-PIE DOOR · DEADMAN · DINING DOME

MASTICATION IS MEDICINE

Galley system — v1.0 in force

GP-IM-169

Revision v1.0 — 23 August 2026

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VETTED BATCH 20 — PASS

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 169

PHASE 169: THE CHEWING DOCTRINE — MOUTH-FEED DISPENSERS, DINING DOMES & PORTION CASSETTES — STATUS: CURRENT — PASS; ONE WORDING PRECISION CORRECTION CARRIED INLINE (VET BATCH 20). No science text talks about eating; this one does. Phase 169 is the serving-and-eating phase Phase 168 flagged: the fleet does not paste-ify its food, because teeth, gums and jawbone are use-it-or-lose-it tissue and children growing dentition need resistance to develop it — chew or lose them. The hardware family follows: mouth-feed dispensers on the battery pepper-grinder pattern (mouth-sized opening, metered feed, button-dosed — auger feed for rice and broccoli, screw-back plunger for the softs); a four-pie radial door that opens outward and clears the bore completely so food passes unsqueezed; a deadman sequence — finger off, screw reverses slightly, then the door closes, nothing crushed, nothing escaping; dining domes, table and four chairs inside clear plastic, so the residue stays in the dome and crumbs never reach the corridor; and spring-loaded portion cassettes dispensing mouth-sized pieces one at a time. The vet passed it with one wording precision carried inline.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-169, v1.0 — 23 August 2026. The one hundred and seventieth manual of the program — 170 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the eating law as seated (contents line, the bodies — the doctrine, the dispenser, the dining room, the cassettes — the physics note and provenance, verbatim) — §2 why chewing is medicine (graded) — §3 the system at the mouth — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 169: **The Chewing Doctrine — Mouth-Feed Dispensers, Dining Domes & Portion Cassettes.** Real texture in zero-G: pepper-grinder-pattern mouth feeders with four-pie

radial doors and deadman sequential close (auger for rice/broccoli, plunger for softs), clear dining domes for residue containment, spring cassettes dispensing mouth-sized solids. Paste rejected as crew-health failure.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

Live instruction, 29 July 2026 — the promised serving-and-eating phase, closing the flag set in Phase 168. No science text talks about eating; this one does. The fleet does not paste-ify its food: people need to chew — especially children — and the hardware exists to serve real texture in zero-G, one mouthful at a time.

THE DOCTRINE (VERBATIM)

- **Chew or Lose Them:** we need to chew — teeth, gums, and jawbone are use-it-or-lose-it tissue, and children growing dentition need resistance to develop it. The fleet cannot convert everything to mush. Paste is the failure mode, not the menu.

THE MOUTH-FEED DISPENSER (VERBATIM)

- **The Pattern:** the battery-operated handheld pepper grinder — a comfortable, mouth-sized opening, metered feed, button-dosed. Scaled to dinner.
- **Two Feed Types, One Body:** screw-feed auger for some foods (rice, broccoli), screw-back plunger for others (mashed potato and the softs) — same device family; only the door mechanism and feed train change.
- **The Four-Pie Door:** behind the lip-stop ridge, a ring actuator opens the door in four pie sections — outward, perpendicular to the cylinder — so the bore clears completely and food passes unsqueezed. The pie pieces run rubber-type seals.
- **The Deadman Sequence:** press the button — food feeds through. Finger off the button — screw and door both stop and close, and they are sequential: the screw reverses slightly first, then the door closes. Nothing is crushed in the seal; nothing escapes unattended.

THE DINING ROOM (VERBATIM)

- **Dining Domes:** table and four chairs inside clear plastic domes — the residue stays in the dome, the crew eats together in the open, and crumbs never reach the corridor.

THE CASSETTES (VERBATIM)

- **Spring-Loaded Portion Cassettes:** rectangular cassettes dispensing small mouth-sized biscuits, mouth-size pieces of cheesecake — portioned solid food, one piece at a time. The list of devices in this family would take all night; the pattern is set.

PHYSICS NOTE — PHASE 169 (KIMI AUDIT: AFFIRMED — MASTICATION IS MEDICINE) (VERBATIM)

- **The chewing doctrine is correct physiology.** Teeth and the periodontal ligament need occlusal load to stay healthy; the jawbone remodels under chewing stress as the long bones remodel under gravity load — the same use-it-or-lose-it law as Phases 99/100's bone protocol, applied to the skull. In children, chew resistance drives jaw development and dental arch formation. A century-ship raised on paste would arrive with a generation of bad jaws. 'The kids need to have teeth' is pediatric medicine, stated plainly. *[WORDING FIXED 15 AUGUST 2026 — vet batch 20, the author's order 'fix the rest'; original: 'exactly as the long bones remodel under gravity load — the tissues and loading patterns are not literally identical; 'exactly' removed']*
- **The pepper-grinder precedent is proven metering.** Battery pepper mills are exactly this: handheld auger metering with button dosing, working at any orientation. Scaling the pattern to a mouth-feeder keeps the one property zero-G demands — food moves only when commanded, by mechanism, never by gravity.
- **The four-pie door is the right aperture engineering.** A camera-iris-style closure squeezes product as it closes; wedges retracting radially outward clear the full bore instantly — broccoli passes whole. The sequential close (screw reverses slightly, then door) pulls product back off the seal line before the rubber seals meet, so the seal never grinds food and never leaks. Deadman-default-closed is the galley clean-seal law at the mouth end: the last centimetre of the food chain obeys the same rule as the first.
- **The dome is the crumb law made architecture.** Phase 168 sealed the transfer; the dome seals the meal — residue contained at the table, vented through filtration, never loosed into the corridor's air. Clear walls because dining is social; containment that isolates the crew from each other would fail the morale mission the whole phase serves.
- **The cassette is the PEZ doctrine, and it is exactly right.** A spring follower behind a rectangular stack dispenses one portion at a time at any orientation — the same positive-displacement family as the plunger container, the silo follower, and the auger: the entire food chain from silo to mouth is one doctrine repeated at five scales. Spring, screw, plunger — nothing in this galley waits for gravity.

ORIGIN OF THOUGHT — PHASE 169 (VERBATIM)

Archer, 29 July 2026: "ok so what no science text talks about is eating, we need to chew, especially children, so we cannot convert everything to mush. ever seen a battery operated hand held pepper

grinder. so we have a mouth sized comfortable opening, screw feed auger style for some, screw back plunger fed for others, same thing, door mech changes, behind the lip stop ridge there is simply a ring actuator that opens the door in four pie sections outward perpendicular to the cylinder, food goes through, with button push when your finger is off the button screw and door both close and stop. they are sequential, the screw reverses slight then the door closes. the pie pieces run the rubber type seals, there are both auger types for rice and broccoli and mashed potato would just be plunger etc etc. dining rooms have table and 4 chairs in clear plastic domes for residue. spring loaded rectangular cassettes can dispense small mouth sized biscuits, mouth size pieces of cheesecake the list of devices would take all night but you get the idea, fck the paste, my kids need to have teeth and our gums needs work also."

Why it matters, in plain English: every space program in history treated food as fuel — squeeze it, swallow it, back to work. A century-ship cannot: it is raising children who will never see Earth, and their jaws, teeth, and gums are being built by what they chew. This phase draws the line the textbooks skipped — texture is medicine, dinner is social, and neither surrenders to zero-G. The devices that follow are all one idea in working clothes: food moves by spring, screw, and plunger, exactly one mouthful at a time, behind doors that close themselves. The fleet will arrive with its teeth. 'Fck the paste' is now formal doctrine, and it is the most humane sentence in the file.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE CHEWING DOCTRINE IS CORRECT PHYSIOLOGY (the phase's own physics note, affirmed): teeth and the periodontal ligament need occlusal load to stay healthy; the jawbone remodels under chewing stress as the long bones remodel under gravity load — the same use-it-or-lose-it law as the bone protocol. Paste is the failure mode, not the menu.

THE PEPPER-GRINDER PRECEDENT IS PROVEN METERING: battery pepper mills are exactly this — handheld auger metering with button dosing, working at any orientation. Food moves only when the mechanism moves it: the one property zero-G demands.

THE FOUR-PIE DOOR IS THE RIGHT APERTURE ENGINEERING: a camera-iris closure squeezes product as it closes; wedges retracting radially outward clear the full bore instantly — broccoli passes whole. The sequential close (screw reverses slightly, then door) means nothing is crushed in the seal.

THE CASSETTE IS THE PEZ DOCTRINE, AND IT IS EXACTLY RIGHT: a spring follower behind a rectangular stack dispenses one portion at a time at any orientation — the same positive-displacement family as the plunger container, the silo follower, and the auger.

§3 — THE SYSTEM AT THE MOUTH

- THE DISPENSER: mouth-sized comfortable opening; battery handheld; button-dosed.

- THE FEEDS: screw auger for rice and broccoli; screw-back plunger for mashed potato and the softs — one body family.
- THE DOOR: four pie sections opening outward, perpendicular to the cylinder, behind a lip-stop ridge; rubber-type seals; bore clears completely.
- THE DEADMAN: finger off — screw stops and reverses slightly, then the door closes; sequential, nothing crushed, nothing escapes.
- THE DOME: table and four chairs inside clear plastic — residue contained, crew together, crumbs never reach the corridor.
- THE CASSETTES: spring-loaded rectangular stacks — mouth-sized biscuits, cheesecake pieces — one portion at a time.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE CHEW-OR-LOSE-THEM DOCTRINE: mastication is medicine; the fleet cannot convert everything to mush.
- THE DEADMAN DOCTRINE: nothing feeds unless held; release stops and seals — the same law as the fleet's other deadman interlocks.
- THE CRUMB LAW MADE ARCHITECTURE: Phase 168 sealed the transfer; the dome seals the meal — containment that is also a room.
- THE POSITIVE-DISPLACEMENT FAMILY: auger, plunger, follower — one doctrine feeding the crew at every scale.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 20 (15 August 2026 — phases 161-175, cross-checked in sitting 307), SEATED. The grade, verbatim from the batch: '169 → PASS / one wording precision correction.' The precision is carried inline in §1; no red, no yellow.

§6 — BUILD SEQUENCE

- STEP 1 — Tool the dispenser body: mouthpiece ergonomics, auger and plunger feed trains interchangeable.
- STEP 2 — Bench the four-pie door: open-clear and seal cycles, crumb intrusion, seal life.
- STEP 3 — Bench the deadman: release-at-any-point stops and seals; worst-case food in the bore.
- STEP 4 — Bench the dome: crumb capture, filtration venting, clean-down time.

- STEP 5 — Bench the cassettes: one-piece-at-a-time across orientations; publish the family drawings, open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 168 — the galley transfer and steam-cook protocol (GP-IM-168): the cook end of the same run.
- PHASE 170 — magnetic tableware and the zero-G wine glass (GP-IM-170): the table end of the same run.
- PHASE 167 — the universal plunger container (GP-IM-167): the dispenser feeds are the plunger doctrine at mouth scale.
- PHASE 99/100 — the bone protocol: the same use-it-or-lose-it law, here applied to the jaw.

§8 — DO-NOT-BUILD

- DO NOT paste-ify the menu — texture is a medical input, not a luxury.
- DO NOT iris-squeeze the food — the door opens outward and clears the bore; crushed seals leak.
- DO NOT feed without the deadman — held to run, released to seal.
- DO NOT eat in the corridor — the dome is the crumb law; the corridor's air is not a bin.

§9 — OWED

OWED: feed-train life data per food class; dome filtration sizing against crumb load; cassette portion mass tolerances (bench, §6). The vet's wording precision is carried inline in §1. The 15 August blanket wording kills are carried inline in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-169, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The one hundred and seventieth manual of the program — 170 of 290. Built 23 August 2026, second of the 20-manual 'ok another 20 lets keep this train rollin' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders

(seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587 and 590): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin'. The phase's own chain, all seated in the master: the contents line and the four bodies (as seated); the live instruction of 29 July 2026 — the promised serving-and-eating phase, closing the flag Phase 168 set; the vet batch 20 grade (PASS with one wording precision, carried inline) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the bench life data, or his word, or his word.

END OF MANUAL — PHASE 169